

FootprintBlend: Selecting Where Object-Removal Outputs Replace the Source

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Abstract

An object-removal model produces a candidate RGB image, but a compositor still decides where that output replaces the source. Full-frame replacement retains intended out-of-mask removal edits but imports unrelated regeneration, whereas naive paste protects source pixels but can undo those edits by restoring removed shadows and reflections. We formalize this spatial arbitration as *return-support selection*. FootprintBlend is a training-free return rule requiring only the source, exact conditioning mask, and one aligned remover RGB. It normalizes far-field additive drift, selects distance-gated strong changes, propagates weaker changes through touching components, and composites at source resolution while keeping pixels outside the soft blend support source-exact. Across five image removers, exact-mask evaluation on OBER and SF-Removal reduces effect-region LPIPS from 0.395 to 0.283 and from 0.295 to 0.252 relative to naive paste; common exact-mask Boundary-B8 LPIPS also falls. Across all 15 dataset-remover pairs on three image datasets, unchanged-region PSNR improves by 1.52–11.55 dB relative to full-output return. These results support a practical one-call RGB return interface while making its effect-coverage and collateral-acceptance trade-off explicit.

1 Introduction

An object-removal output is a candidate reconstruction, not automatically the image that should be delivered. It can contain the intended removal near the target while also changing unrelated texture, color, or resolution elsewhere. The final compositor must therefore decide where output pixels replace source pixels and where the original image remains authoritative (Figure 1).

Two common policies expose the problem. Returning the full output retains the remover’s edits to shadows, reflections, and color spill beyond the user mask, but also accepts every collateral model change. Naive paste makes the opposite choice: it restores the source outside the conditioning mask, protecting unrelated detail while potentially undoing a legitimate out-of-mask removal edit. Neither policy adapts its replacement region to the actual output.

We distinguish this compositing decision from generation. At the *generation interface*, a user marks an object and a model predicts the scene without it. At the *return interface*, that prediction is reconciled with the source. Effect-aware systems increasingly generate changes beyond the supplied

mask (Wei et al. 2025; Zhu et al. 2025; Zhao et al. 2026; Zhou et al. 2026), while black-box services such as FLUX Erase expose only a completed RGB result (Black Forest Labs 2026). Even a successful generator therefore leaves a separate question: *where should its output win over the source?*

We call the selected replacement region the *return support*, and its estimation *return-support selection*. This names a post-inference spatial decision, not a semantic effect mask. Support that is too small restores source pixels over a legitimate edit; support that is too large accepts model changes over untouched background. Moreover, RGB difference alone cannot identify intent: a nearby hallucination and a removed reflection can produce the same residual. We therefore estimate a *source-anchored change support* under explicit locality and connectivity priors and evaluate both effect miss and collateral acceptance.

The selected support may include a transition zone needed to reconcile two RGB images, and it may reject an observed change unrelated to the user request. Overlap with a photometric ground-truth change map measures coverage, but the decision must also be judged by the perceptual cost of accepted and rejected pixels. No single overlap score is therefore a sufficient definition of a correct replacement region.

FootprintBlend implements this return interface from only spatially aligned (I, M, Y) : source RGB, user mask, and remover RGB. It removes robust far-field additive color drift, selects a per-image residual threshold, and grows strong and weak changes near the mask. The hard support is feathered and composited at source resolution. Outside the effective blend support, the in-memory output is exactly the source; native detail is preserved, not regenerated. The same parameters are used throughout without model-specific training, calibration, model internals, or another remover call.

The experiments separate questions that a raw-to-postprocess comparison would otherwise conflate. Full-output return tests source preservation; naive paste and matched compositing controls test whether changing support is necessary. FootprintBlend recovers substantially more effect support on OBER and SF-Removal while retaining far more unchanged content than full-frame return. It also exposes the cost of that choice: expanded support can accept collateral changes, and on RemovalBench the effect metric is tied with naive paste. This is a measured operating point,

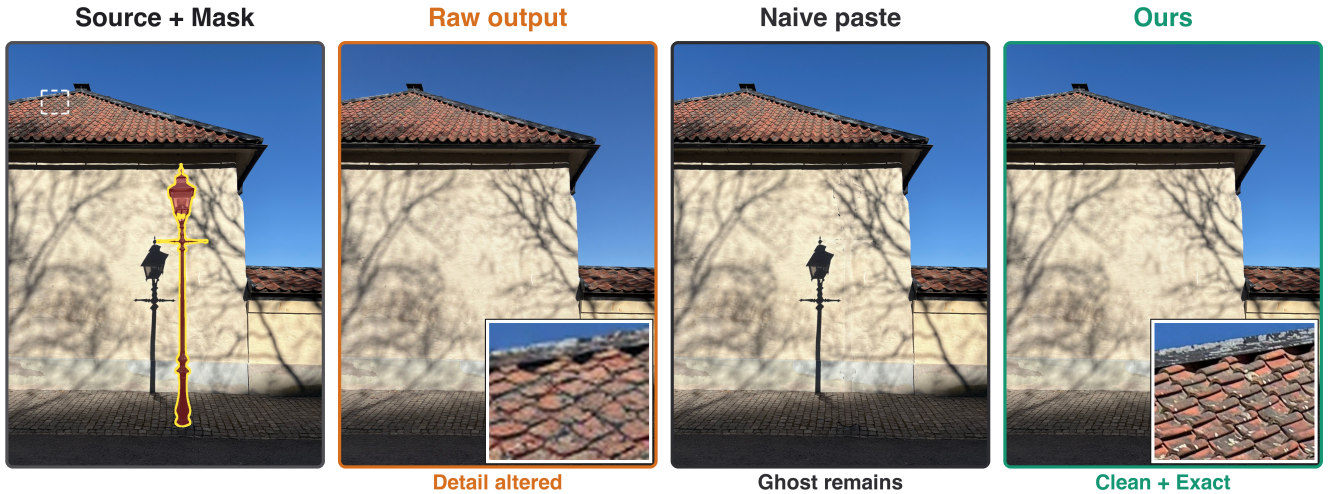


Figure 1: Return policy determines which parts of a completed removal reach the delivered image. From left: the source and conditioning mask; the raw full-frame output, which removes the target lamp but alters unrelated roof texture (inset); naive paste, which restores source pixels outside the mask but leaves a visible out-of-mask ghost; and FootprintBlend, which retains the removal over inferred support while keeping unrelated roof detail source-exact. The dashed box marks the enlarged roof region.

not recovery of a unique causal footprint.

This separation is central to the evidence. Raw-to-FootprintBlend comparisons ask whether selective return protects the source; naive-paste comparisons ask whether accepting changes beyond the user mask preserves legitimate edits; and fixed-support feather and Poisson controls ask whether the result can be explained by seam treatment alone. Together, these comparisons distinguish the value of a return interface from the quality of the underlying remover and from the choice of blending operator.

Our contributions are:

- We identify and formalize the post-inference decision of where an object-removal output should replace its source, with explicit effect-miss and collateral-acceptance failure modes.
- We introduce FootprintBlend, a training-free postprocessor operating on source RGB, the conditioning mask, and final remover RGB. Its native-resolution compositor guarantees source-exact pixels outside its effective blend support.
- We isolate support selection from seam treatment and evaluate one shared operating point across five image removers and three datasets, reporting both its gains and its collateral-acceptance cost.

2 Related Work

Where Edit Support Comes From

Object removal has long been framed as masked image completion, from classical exemplar-based inpainting (?) to modern generative models (Pathak et al. 2016; Yu et al. 2019; Lugmayr et al. 2022; Rombach et al. 2022; Podell et al. 2024). Modern systems increasingly model shadows, reflections, and other out-of-mask effects. Image methods obtain

that support from supervision, geometry, attention, or learned alpha prediction (Winter et al. 2024; Wei et al. 2025; Zhu et al. 2025; Zhao et al. 2026; Yu et al. 2025; Zhou et al. 2026; Yang et al. 2025). These methods derive support from model- or training-specific signals. ObjectClear’s attention-guided fusion and OSOR’s alpha head are especially close in purpose, but their internal signals are not directly available when a caller holds only the source, conditioning mask, and completed RGB output. FootprintBlend operates after inference under exactly that access constraint.

The distinction is one of access rather than desired semantics. Supervised effect masks can teach a model what shadows or reflections look like; geometry can connect an object to causal scene changes; attention and alpha heads can expose where a particular network acted. Such signals may provide stronger semantic evidence than RGB differencing, but they depend on training data, architecture, weights, or inference state and are not exposed by the final RGB alone. An external estimator can still be layered onto a closed API or cached output. We study one such restricted post-inference contract available in our setting: one source, one exact conditioning mask, and one aligned RGB result. Our claim is therefore not that output difference supersedes model-internal effect reasoning, but that a return rule remains necessary when those signals are unavailable.

Post-Hoc Change Localization

Several neighboring tasks localize image changes without solving our return decision. Omnimate associates objects with their effects through per-video optimization (Lu et al. 2021), and instance-shadow systems learn object-shadow associations from supervision (Wang et al. 2020). Generic change detection compares scenes under viewpoint and illumination variation (Sachdeva and Zisserman 2023), while

manipulation localization infers edited regions from forensic traces in a single image (Guillaro et al. 2023). These methods seek semantic associations, scene changes, or evidence of manipulation. A return rule instead receives the intervention mask and must decide which differences in a specific remover output should survive source restoration.

Classical seeded region growing and hysteresis (Adams and Bischof 1994; Canny 1986) provide useful locality and connectivity priors, but do not define the evidence, operating point, or return contract. Direct thresholding is also insufficient when a full-frame generator introduces small channel offsets across nominally unchanged pixels. FootprintBlend combines these old spatial primitives with drift-normalized, image-adaptive change evidence. The contribution lies in the black-box decision formulation and its source-exact return behavior, not in presenting region growing as a new segmentation algorithm.

Locality, Compositing, and Access Level

Editing locality can be imposed inside a sampler: Blended Diffusion (Avrahami, Lischinski, and Fried 2022; Avrahami, Fried, and Lischinski 2023) mixes under a supplied mask, and Differential Diffusion (Levin and Fried 2025) accepts spatial edit strength. Learned refiners and latent compositors use still deeper access (Zheng et al. 2026, 2025; Bradbury and Zhong 2026), while SeamEdit (Lyu and Lei 2026) uses repeated black-box calls and candidate fusion. Our setting allows none of these: one aligned RGB result, no weights, latent state, attention, or additional call.

This access boundary changes what can be guaranteed. Sampler-level methods may prevent changes before they are generated, and repeated-call methods may search over candidates. A post-hoc return rule cannot repair an in-mask hallucination or recover an out-of-mask edit that a strictly clamped interface never emitted. It can only arbitrate observed pixels in Y . In exchange, the same rule can be applied to heterogeneous diffusion, one-step, and specialist removal outputs without retraining or model-specific instrumentation. This complementarity is why we treat generation and return as separate interfaces rather than competing model families.

Once support is known, blending is classical. Alpha and feathering (Porter and Duff 1984), multiband blending (Burt and Adelson 1983), Poisson editing (Perez, Gangnet, and Blake 2003), mean-value coordinates (Farbman et al. 2009), guided filtering (He, Sun, and Tang 2010), and learned harmonization (Tsai et al. 2017; Ke et al. 2022; Zhou et al. 2024) all address appearance at a supplied boundary. They do not decide whether an out-of-mask shadow deletion should be retained. We therefore separate *support selection* from *blend treatment* and use these operators as controls, rather than presenting classical region growing (Adams and Bischof 1994; Canny 1986) or alpha blending as new primitives.

Evaluation of Removal and Return

Paired benchmarks include RemovalBench (Wei et al. 2025) and OBER (Zhao et al. 2026). Reference-free metrics such as ReMOVE (Chandrasekar et al. 2024) and ReMOVE+ (Zhao et al. 2026) score erasure without a paired target. Our central estimand differs from model ranking: for a fixed remover

output, how does the return rule change effect-region fidelity and unchanged-region fidelity? Region-decomposed paired metrics and support diagnostics make those opposing costs visible.

This estimand requires stricter controls than a conventional model leaderboard. Raw and returned images must share the same generated RGB, otherwise generation quality is confounded with compositing. All strategies must use the exact mask given to the remover, otherwise an eroded paste boundary creates an artificial miss. Boundary comparisons must use a common support when ranking strategies, and uncertainty must be paired within each fixed remover output. Our protocol follows these constraints. The result is not a claim that FootprintBlend improves the underlying generator; it measures how much of a fixed generation should be returned to the source image.

3 Return-Support Formulation and Method Problem and Return Contract

The postprocessor receives a source image $I \in \{0, \dots, 255\}^{H \times W \times 3}$, the exact binary mask M supplied to the remover, and its spatially aligned RGB output Y . A return rule chooses an effective support C on which Y may contribute. Too little support causes *effect miss*: source pixels overwrite a legitimate removal. Too much support causes *collateral acceptance*: unrelated model changes overwrite the source. The latent semantic set separating these cases is unobserved and, from (I, M, Y) alone, generally unidentifiable. FootprintBlend implements one fixed operating point under a source-change prior: accepted changes should be strong relative to far-field noise and spatially related to the requested mask.

Drift-Normalized Change Evidence

Footprint inference runs on an analysis grid whose long side is at most 1024 pixels; compositing runs at (H, W) . Y is aligned by Lanczos-4 when required, and masks use nearest-neighbor resizing. After radius-5 Gaussian smoothing, the signed residual is

$$\delta_c(x) = B_5(Y_c)(x) - B_5(I_c)(x), \quad (1)$$

and remains signed until its robust per-channel offset is removed.

Let Ω_a be the analysis grid, let $D_M(x)$ be its Euclidean distance transform from M in analysis pixels, let ℓ_a be its diagonal, and set $r = 100\ell_a/1448$. The nominal far field is

$$\mathcal{F}_0 = \{x : D_M(x) > \max(0.25\ell_a, 3r)\}. \quad (2)$$

Statistics use the grid stride $s_g = \max\{1, \lceil \sqrt{|\Omega_a|/65536} \rceil\}$. Let Ω_s denote that strided grid and $q = \min\{\max(256, \lfloor 0.01|\Omega_s| \rfloor), |\Omega_s|\}$. If fewer than q pixels lie in $\mathcal{F}_s = \Omega_s \cap \mathcal{F}_0$, the q grid points farthest from M replace $\mathcal{F}_s$. This fallback guarantees the sample count; the quarter-diagonal exclusion remains a spatial prior, not a semantic guarantee. The selected samples define the channel offset; rectification is then applied across Ω_a :

$$\hat{o}_c = \text{median}_{x \in \mathcal{F}_s} \delta_c(x), \quad d(x) = \max_c |\delta_c(x) - \hat{o}_c|. \quad (3)$$

Thus constant per-channel drift is removed before absolute-value rectification.

Robust Threshold and Mask-Gated Growth

On the normalized far-field samples, define

$$m = \operatorname{median}_{x \in \mathcal{F}_s} d(x), \quad (4)$$

$$a = \operatorname{median}_{x \in \mathcal{F}_s} |d(x) - m|, \quad (5)$$

$$t = \min\{7.0, \max\{1.0, m + 4a\}\}. \quad (6)$$

This is robust threshold selection, with no false-positive-rate interpretation. All experiments use one fixed parameter set: $k = 4$, floor 1.0, and cap 7.0; none is retuned by model, image, or dataset.

The source-gradient penalty is computed after threshold selection. OpenCV RGB grayscale conversion is followed by 3×3 Sobel derivatives. If G is their magnitude, the implementation forms

$$g(x) = \frac{G(x)}{\max_{z \in \Omega_a} G(z) + 10^{-5}}, \quad (7)$$

$$p(x) = d(x) \exp(-3 \mathcal{D}_{3 \times 3}(g)(x)), \quad (8)$$

where $\mathcal{D}_{3 \times 3}$ is one dilation with an all-ones kernel.

Round zero uses $D_0 = D_M$; each later round measures distance from the preceding round’s accepted candidate map. Every round evaluates

$$e_j(x) = \begin{cases} p(x) \exp(-D_j(x)/r), & D_j(x) < 3r, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Round zero applies the strict test $e_0 > t$ and accepts every passing pixel, including disconnected strong islands. At most two weak rounds follow, each using $e_j > 0.6t$. In a weak round, 8-connected components are computed on the union of the current support and the thresholded candidates; only components intersecting the current support contribute new pixels. Growth stops as soon as a round adds none.

The accumulated support then undergoes binary closing followed by binary opening with the radius-2 Euclidean disk $\{(u, v) : u^2 + v^2 \leq 4\}$, and M is unioned back. This sequence does not impose a final connectedness guarantee. When analysis was downsampled, the resulting hard footprint F is returned to (H, W) by nearest-neighbor interpolation.

Source-Exact Native Return

Let $L = \max(H, W)$ and $f = \max\{1, \lfloor 16L/1024 \rfloor\}$ native pixels. The implementation crops the bounding box of F with padding $f + 5$. If that patch has long side above 1024, its hard mask is resized by nearest neighbor with scale $\rho = 1024/L_{\text{patch}}$; otherwise $\rho = 1$. The feather used for the exterior Euclidean distance transform is $f_a = \max\{1, \rho f\}$, and $u(x) = D_F(x)/f_a$ at that patch scale. The alpha is

$$\alpha(x) = \begin{cases} 1, & x \in F, \\ \frac{1}{2} (1 + \cos(\pi u(x))), & 0 < u(x) \leq 1, \\ 0, & u(x) > 1. \end{cases} \quad (10)$$

When computed on the capped patch, α is bilinearly resized to the native patch and reset to one on F .

The final blend is evaluated in floating point at native resolution,

$$O(x) = \alpha(x)Y(x) + (1 - \alpha(x))I(x). \quad (11)$$

Values are clamped to $[0, 255]$ and cast to 8-bit RGB. The result buffer starts as an exact copy of I , so pixels outside the implementation blend support $C_0 = \{x : \alpha(x) > 0\}$ remain byte-identical to the source. Footprint metrics evaluate F ; Y can receive nonzero weight in the feather $C_0 \setminus F$, and the realized byte-change support $\{x : O(x) \neq I(x)\}$ is a subset of C_0 .

Identifiability and Applicability

FootprintBlend estimates source-anchored change support; RGB alone cannot separate an intended out-of-mask edit from a nearby hallucination. Its far-field normalization removes constant per-channel additive drift, but does not correct gains, tone curves, spatially varying illumination, crop changes, or geometric misregistration. Preserving removed shadows, reflections, and spill outside M also requires the remover to expose them in its RGB difference. For a byte-exact mask-clamped output, inferred support over pixels where $Y = I$ is operationally inert. These are interface limits, not failures that additional threshold tuning can remove.

4 Experimental Setup

Datasets

Image evaluation uses RemovalBench-69 (Wei et al. 2025), OBER-Test-163 (Zhao et al. 2026), and SF-Removal-sub (Li et al. 2026). RemovalBench provides development and mechanism evidence; OBER-Test and SF-Removal were not used to tune the postprocessor and provide confirmatory raw-to-plugin comparisons. Footprint ground truth is the predefined photometric change support whose maximum source-to-target channel difference exceeds $8/255$. All reported sample counts refer to source images, and every comparison is paired within the same remover output.

Evaluated Removers

The five image removers are ObjectClear (Zhao et al. 2026), GeoRemover (Zhu et al. 2025), OmniEraser (Wei et al. 2025), OSOR (Zhou et al. 2026), and SDXL-Inpainting (Podell et al. 2024). These systems span specialist effect-aware removers and a general inpainting baseline. Each receives the same FootprintBlend parameters, with no model-specific tuning; all quantitative comparisons are within remover rather than cross-model rankings.

Metrics and Protocol

Full-reference quality uses PSNR, SSIM, and LPIPS over four predefined regions: the full frame, the effect union, background only, and an 8-pixel boundary band. All internal-review results use the exact conditioning mask. Footprint targets use the predefined photometric change support $\max_c |I_c - I_c^{\text{GT}}| > 8/255$. Footprint recall is the sole count metric and measures missed target change. Effect-region LPIPS measures the consequence of under-coverage, while FP-region LPIPS measures fidelity where an estimated footprint accepts remover pixels over GT-unchanged content. Headline boundary comparisons use one common 8-pixel band around the exact conditioning mask for every strategy; strategy-specific final-footprint bands are diagnostic only.

The central comparisons are paired within a fixed remover output. Raw versus FootprintBlend tests how much source fidelity the return rule recovers. Naive paste versus FootprintBlend tests the cost and benefit of expanding support beyond the exact conditioning mask. BG LPIPS and Effect LPIPS expose the two sides of this trade-off; BG PSNR and Full LPIPS are secondary views. Paired 95% intervals use 10,000 cluster-bootstrap resamples with seed 20260720, with source images as clusters. No cross-model significance claim is made.

The main compositing controls are naive paste, Gaussian feathering, and Poisson normal editing (Perez, Gangnet, and Blake 2003). All controls receive the same exact mask and generated RGB; the comparison therefore changes only return support or blend treatment.

Deployment Characteristics

At deployment, FootprintBlend is a deterministic post-inference component requiring no learned weights, model-specific training or calibration, model instrumentation, or additional remover call. The analysis stage caps its long side at 1024 pixels, while final compositing is performed over a padded native-resolution crop. The returned buffer starts as an exact copy of the source, providing a directly testable invariant: every pixel outside the effective soft support remains byte-identical to its input.

Runtime separates capped-domain footprint inference from the resolution-proportional blend. On one pinned hardware thread, footprint inference measured 61.17–232.94 ms median and the separate native-resolution CPU blend measured 5.33–341.24 ms median over real 0.262–25.082 MP inputs. A byte-exact CUDA implementation accelerates the isolated blend stage by $2.854\text{--}9.083\times$. These measurements describe postprocessing only and exclude remover inference.

For integration, the required contract is explicit: source RGB and mask must share native geometry, remover RGB must be spatially aligned to that source, and the exact mask sent to the remover must also be sent to FootprintBlend. The implementation should reject or route geometry mismatches rather than silently resizing an unknown crop. A conservative deployment can retain naive paste as a fallback for mask-clamped outputs and interfaces that cannot provide alignment guarantees.

The method also produces useful operational diagnostics without exposing model internals: the estimated hard footprint, soft blend support, far-field channel offset, calibrated threshold, and accepted-support fraction. These quantities make unusually large supports or weak far-field evidence observable at the return layer. They do not certify semantic correctness, but they provide actionable monitoring signals and allow the return policy to be tested independently of the underlying remover.

5 Results

Returning the Edit without Returning the Frame

The first question is whether selective return improves fidelity over returning the complete remover frame when the generated RGB is held fixed. Table 1 compares these return

policies under one exact-mask protocol for every dataset–remover combination. Reporting all 15 paired comparisons tests whether an aggregate gain is broad or driven by a small subset of removers. The deltas isolate return policy; they neither compare generator quality nor rank removers.

Every one of the 15 comparisons moves all four reported metrics in the favorable direction. Background PSNR gains range from 1.52 to 11.55 dB; background and full-frame LPIPS reductions range from 0.003 to 0.199 and 0.003 to 0.196, respectively. Effect LPIPS also falls in every case, by 0.001–0.086. The result is therefore not an artifact of averaging incompatible remover behavior. It establishes source preservation relative to full-output return, but does not yet establish that inferred support is preferable to naive paste.

Figure 2 shows the same return-policy distinction in two sampled frames of a video removal. The inferred support adapts to each frame while preserving distant mountain and snow detail that changes in the raw output. Because FootprintBlend operates independently per frame and the figure is qualitative, this example supports only spatial source preservation.

Support Selection, Not Seam Smoothing

The next question is whether the boundary gain can be reproduced without changing return support. Table 2 compares FootprintBlend with three exact-mask controls on RemovalBench-69, separately for each remover. The three baseline strategies retain conditioning-mask support; only FootprintBlend selects support beyond that mask.

Table 2 shows that fixed-support seam treatments do not reproduce the support-selection result. Feathering and Poisson blending worsen the common boundary for every remover. FootprintBlend lowers Boundary-8 LPIPS for four of five removers, but SDXL-Inpaint is a clear exception (0.616 to 0.645). The unweighted five-remover mean still falls by 20.3%, from 0.556 to 0.443, with a paired difference of -0.112841 (95% CI $[-0.163294, -0.064326]$).

The regional deltas expose a second heterogeneity hidden by the former mean: GeoRemover improves both Effect and Background LPIPS, OSOR has a small effect gain with a background cost, and the remaining three removers trade higher regional LPIPS for their selected support. At the five-remover level, Effect LPIPS remains statistically tied with naive paste ($+0.000581$, 95% CI $[-0.014136, +0.015539]$). Thus the evidence supports a broad but not universal common-boundary gain, not an effect-region gain on this development set.

Effect Coverage and Accepted-Background Severity

Support expansion can recover target changes missed by naive paste, but it can also retain remover pixels where the ground truth is unchanged. Table 3 asks how expansion changes target-effect coverage and the photometric severity of accepted pixels. Because accepted-background LPIPS is conditional on those pixels, the table does not measure how many are accepted.

On OBER and SF-Removal, FootprintBlend raises target recall from 0.584 to 0.864 and from 0.466 to 0.722, respectively, while lowering Effect, accepted-background,

Dataset	Remover	Δ BG PSNR $\uparrow$	Δ BG LPIPS $\downarrow$	Δ Full LPIPS $\downarrow$	Δ Effect LPIPS $\downarrow$
OBER-Test	ObjectClear	+1.516 [†]	−0.003 [†]	−0.003 [†]	−0.001 [†]
	GeoRemover	+7.430 [†]	−0.079 [†]	−0.072 [†]	−0.004 [†]
	OmniEraser	+8.236 [†]	−0.095 [†]	−0.087 [†]	−0.006 [†]
	OSOR	+9.310 [†]	−0.052 [†]	−0.048 [†]	−0.005 [†]
	SDXL-Inpaint	+11.551 [†]	−0.036 [†]	−0.034 [†]	−0.005 [†]
RemovalBench	ObjectClear	+2.975 [†]	−0.144 [†]	−0.140 [†]	−0.050 [†]
	GeoRemover	+6.335 [†]	−0.121 [†]	−0.117 [†]	−0.036 [†]
	OmniEraser	+6.537 [†]	−0.153 [†]	−0.148 [†]	−0.049 [†]
	OSOR	+7.044 [†]	−0.199 [†]	−0.196 [†]	−0.086 [†]
	SDXL-Inpaint	+9.824 [†]	−0.111 [†]	−0.109 [†]	−0.048 [†]
SF-Removal-sub	ObjectClear	+5.001 [†]	−0.028 [†]	−0.026 [†]	−0.004 [†]
	GeoRemover	+11.207 [†]	−0.050 [†]	−0.048 [†]	−0.014 [†]
	OmniEraser	+8.758 [†]	−0.084 [†]	−0.081 [†]	−0.011 [†]
	OSOR	+8.181 [†]	−0.049 [†]	−0.047 [†]	−0.010 [†]
	SDXL-Inpaint	+9.788 [†]	−0.031 [†]	−0.030 [†]	−0.006 [†]

Table 1: Per-remover change from raw full-output return to FootprintBlend using the exact conditioning mask: $\Delta = \text{FootprintBlend} - \text{raw}$. Positive PSNR and negative LPIPS deltas are favorable. Both conditions share the same generated RGB. A dagger denotes a paired 95% source-image-cluster bootstrap interval excluding zero; all displayed differences satisfy this criterion.

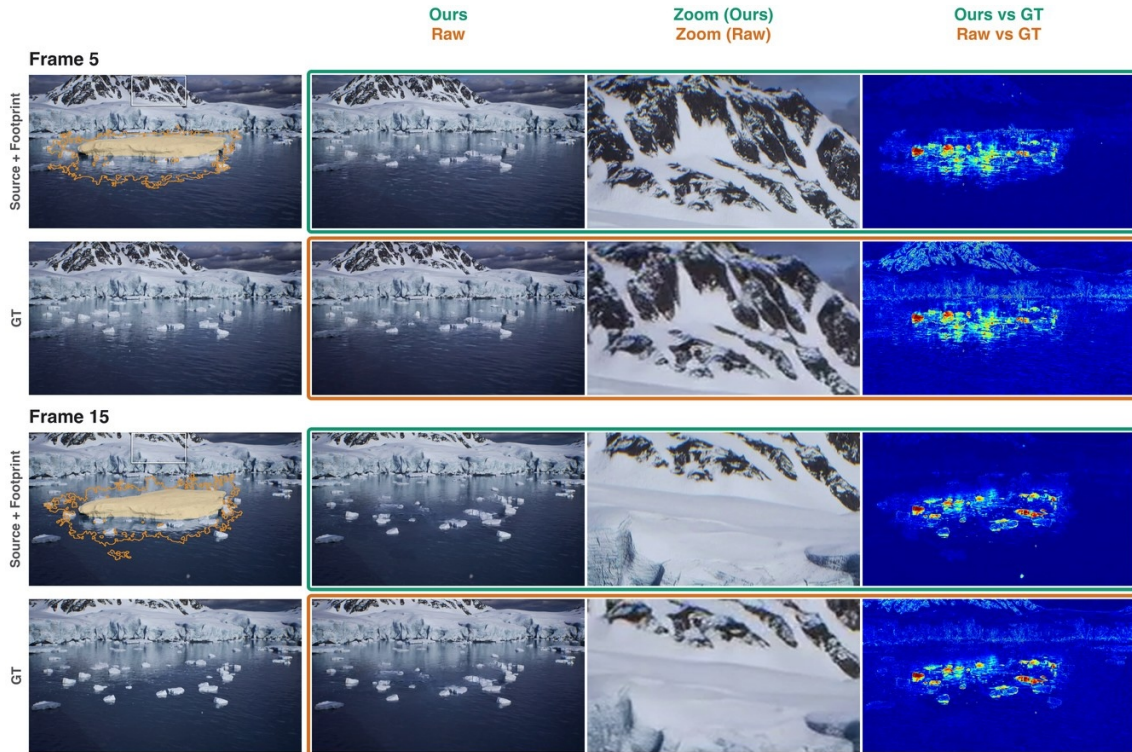


Figure 2: Qualitative source preservation in sampled video frames. Frames 5 and 15 of a ROSE sequence (Miao et al. 2025) compare FootprintBlend (green) with raw return (orange). Source panels overlay the inferred hard footprint; zooms inspect distant background; and heatmaps show absolute RGB error to ground truth, with warmer colors indicating larger error. FootprintBlend preserves remote detail and localizes the returned change. The figure assesses spatial return behavior, not temporal coherence.

and Boundary-8 LPIPS. Lower accepted-background LPIPS means the retained GT-unchanged pixels are less photo-metrically severe; it does not imply that fewer such pix-

els are retained. On RemovalBench-69, recall rises from 0.258 to 0.454 and Boundary-8 LPIPS falls from 0.556 to 0.443, but Effect LPIPS is statistically tied with naive paste

Remover	Common Boundary-8 LPIPS ↓				FootprintBlend – naive paste	
	Naive paste	Feather	Poisson	FootprintBlend	Δ Effect ↓	Δ BG ↓
ObjectClear	0.559	0.573	0.633	0.416	+0.014	+0.025
GeoRemover	0.516	0.543	0.610	0.324	−0.027	−0.002
OmniEraser	0.524	0.554	0.610	0.394	+0.010	+0.020
OSOR	0.563	0.582	0.639	0.435	−0.003	+0.003
SDXL-Inpaint	0.616	0.623	0.646	0.645	+0.009	+0.001

Table 2: Per-remover exact-mask compositing controls on RemovalBench-69. Boundary-8 uses the same 8-pixel band around the conditioning mask for every strategy. The last two columns report FootprintBlend minus naive paste on predefined effect and background regions; negative values favor FootprintBlend. All entries are LPIPS and are rounded to three decimals; statistical comparisons use unrounded values. Bold marks the lowest displayed common-boundary value in each row.

Metric (OBER / SF / RB)	Naive paste	Feather	FootprintBlend
Target recall ↑	0.584 / 0.466 / 0.258	0.584 / 0.466 / 0.258	0.864 / 0.722 / 0.454
Effect LPIPS ↓	0.395 / 0.295 / 0.348	0.498 / 0.339 / 0.348	0.283 / 0.252 / 0.349
Accepted-BG LPIPS ↓	0.311 / 0.337 / 0.523	0.498 / 0.398 / 0.536	0.186 / 0.150 / 0.376
Boundary-8 LPIPS ↓	0.349 / 0.358 / 0.556	0.522 / 0.426 / 0.575	0.264 / 0.281 / 0.443

Table 3: Effect coverage and accepted-background severity using exact conditioning masks, averaged over the same five removers as Table 1. Values within each cell are ordered OBER/SF/RB. Accepted-BG LPIPS is computed only over accepted GT-unchanged pixels and measures their photometric severity, not their count. Boundary-8 uses one common band around the exact conditioning mask.

(+0.000581, 95% CI [−0.014136, +0.015539]). The table therefore supports a useful fixed operating point, not geometric dominance or recovery of a unique semantic footprint.

Implementation and Integration Checks

The three result blocks support different parts of the internal decision. Table 1 establishes that selective return consistently protects unchanged source content across heterogeneous removers. Table 2 rules out fixed-support feathering or Poisson editing as an explanation for the common-boundary gain. Table 3 then exposes the operational cost of expanded support instead of hiding it inside a full-frame average. Together they support integration as a return-layer component, not a claim that the underlying generators have improved.

Implementation checks target failure modes that aggregate perceptual metrics would miss. In a 30-sample CPU–CUDA equivalence test, all 29,771,264 output pixels and all independently recomputed hard footprints were byte-exact. Separate exact-mask forensics confirmed source-resolution binary masks, found no changed pixels when mask conversion was varied from > 127 to > 0 , and measured reliable exterior-band phase estimates below 0.34 pixels. A one-pixel dilation improved cut-band LPIPS only gradually, by 6.8%–10.1%, rather than eliminating the observed seam. These checks reduce the risk that the measured gain is an artifact of CPU/GPU divergence, mask quantization, or a one-pixel alignment error.

The integration boundary remains narrow. FootprintBlend does not alter remover inference, rank remover quality, or require a second generation. It receives one completed output and returns one native-resolution image plus optional support diagnostics. This separation allows rollout, monitoring, and fallback policy to be evaluated independently for each

upstream remover.

6 Limitations and Failure Modes

Intended effects and collateral hallucinations are not identifiable from aligned RGB difference alone. Locality and connectivity priors can accept a strong nearby hallucination or reject a weak disconnected reflection, so effect recall and collateral acceptance remain a genuine operating trade-off rather than an error that one threshold can eliminate (Table 3). A strictly mask-clamped remover presents the opposite limit: if it never exposes an out-of-mask edit in Y , no post-hoc method observing only (I, M, Y) can recover that edit.

The input must remain spatially co-registered. Far-field normalization handles constant per-channel additive drift, not camera motion, crops, geometric warps, exposure gain, tone curves, or spatially varying illumination. Frame-dominant effects can also contaminate the far-field estimate. Deployment should therefore validate alignment and preserve a fallback to naive paste for unsupported interfaces.

7 Conclusion

Object removal needs a compositor that decides where generated pixels should replace source pixels. FootprintBlend makes that decision from drift-normalized RGB change under explicit spatial priors. Across five image removers, the same operating point preserves substantially more source content than full-frame return and, on effect-heavy datasets, preserves more legitimate effect edits than naive paste. Its collateral-acceptance cost and identifiability limits are equally important: the contribution is a measurable replacement trade-off, not semantic recovery of a unique causal footprint.

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